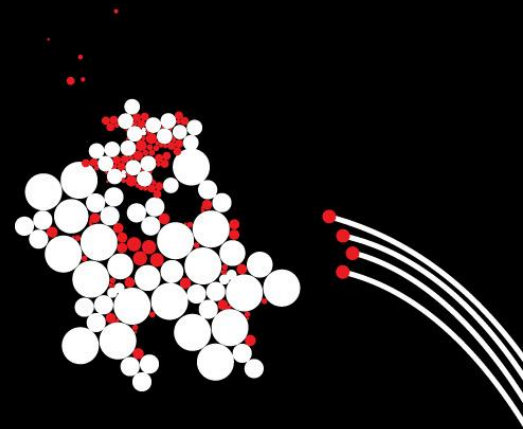


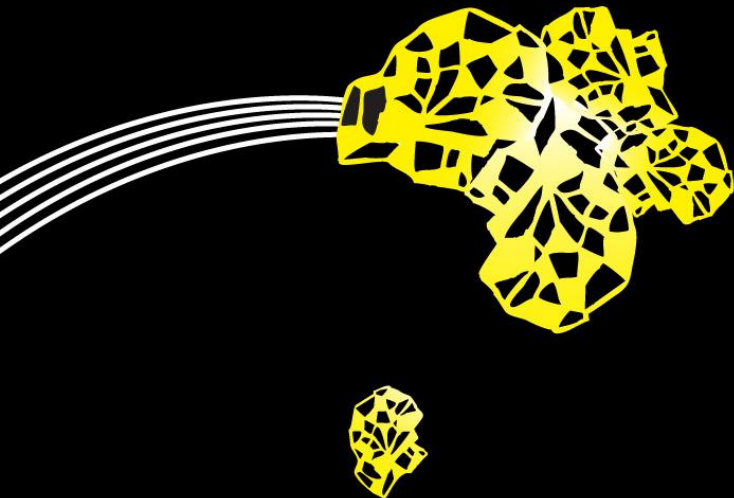
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DS & AI [2202200020] SEEING THROUGH THE HYPE



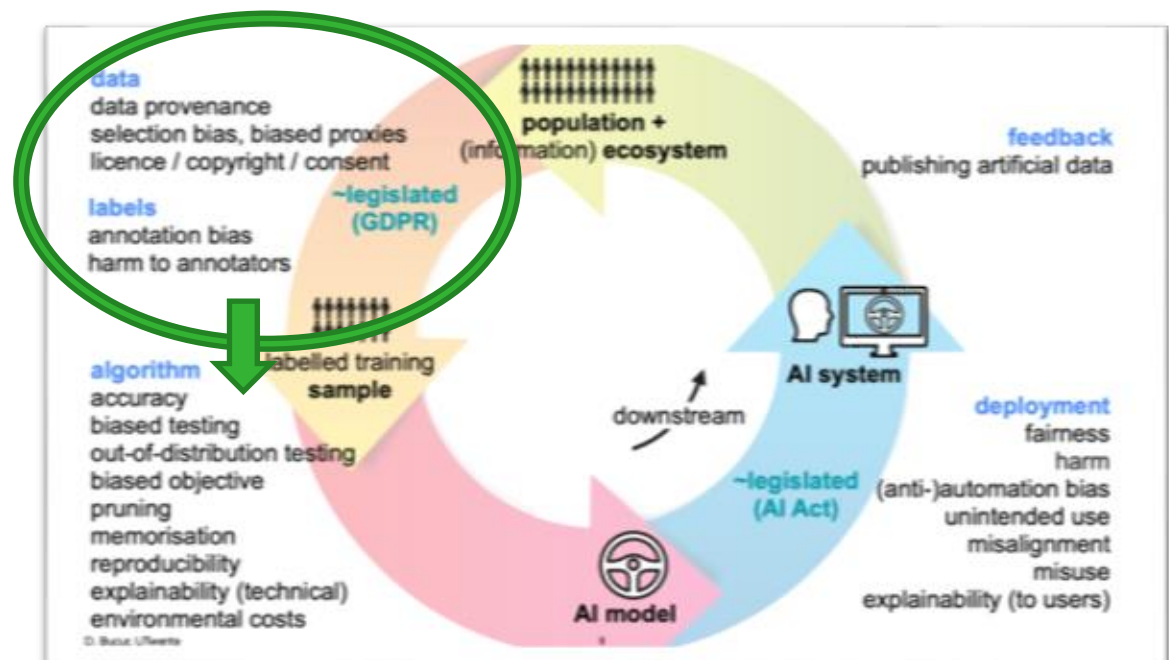
DATA QUALITY – LECTURE 1 MOTIVATION & WHAT IS DQ & NEURO-SYMBOLIC AI

MAURICE VAN KEULEN



THE DQ PART

- DQ: The power of Neuro-Symbolic AI (Lec 1)
- DQ: Sensitivity of ML to DQ problems (Lab 1)
- DQ: Detecting, solving, and mitigating DQ problems (Lec 2+3, Lab 2)



Grade for module part “DQ”
is 70% of grade of
DQ&Ethics study unit

OVERVIEW OF DQ PART

3 Lectures

- Lec 1: What is DQ and Neuro-Symbolic AI?
 - Lab 1: Sensitivity of ML to DQ problems
- Lec 2&3: Detection, cleaning & mitigation of DQ problems
 - Lab 2: Detection & cleaning (mostly on-paper)

DQ assessment

- Test: 80%; Lab assignments 20%
- Lab 1: 30 points; Lab 2: 70 points

STUDY MATERIAL OF *THIS LECTURE*

Book (available as e-book through UT's library)

- Shazia Sadiq ed. (2013). *Handbook of data quality: research and practice*. Heidelberg, Germany: Springer. ISBN 978-3-642-36256-9. DOI [10.1007/978-3-642-36257-6](https://doi.org/10.1007/978-3-642-36257-6)
 - Chp I-1: “Data Quality Management Past, Present, and Future”
 - Chp I-3: “Cost and Value Management for Data Quality”
 - Chp II-1: “Data Warehouse Quality: Summary and Outlook”
(but part of this chapter is also used in Lecture 2)
- Emmanuel, T., et al. A survey on missing data in machine learning. *J Big Data* 8, 140 (2021). DOI 10.1186/s40537-021-00516-9

And the slides of course

- Part of these chapters, I will address in the lecture
... but I expect you to have studied these chapters in full for the exam

RESPONSIBLE USE OF GENAI IS ALLOWED WITH DQ

THINK OF THE PURPOSE (= LEARNING GOALS)

What is responsible use?

- Don't do obvious non-responsible actions such as putting sensitive information in the prompt
- **Purpose** why you sit here is to learn!
Let GenAI help you learn faster & more deeply

Example

Purpose of DQ Lab 1 is to experience yourself how different kinds of DQ problems affect ML performance

- Purpose is not to learn to program, so if you can experiment more with GenAI generated code, then go ahead.
- If your programming skills are not that good yet, then program yourself, ask GenAI to give you improvement suggestions, to explain things to you, etc.

LEARNING GOALS

After this module part,

- students have knowledge of multiple facets of data quality and understand its impact on machine learning and organizations;
- can apply several techniques for detecting data quality problems based on data exploration-driven, constraint-driven, and machine learning-driven approaches;
- can apply several techniques for cleaning or otherwise improving data quality such as deletion, consistent query answering, and database repairs;
- develop their critical attitude towards data to the level of assuming that there will always be quality problems in their data and that they will be prone to start actively searching for them

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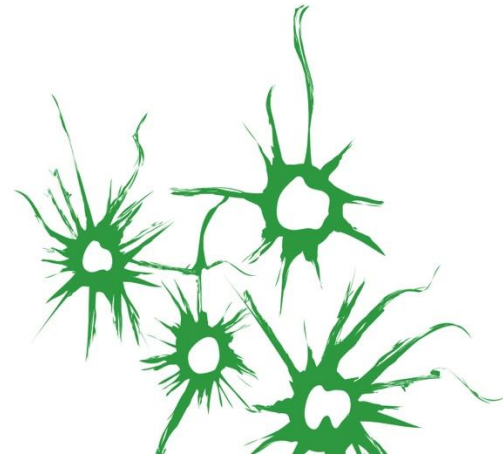
Conference: ETIM 2022

AI IN HEALTH

THE STRUGGLE FOR HIGH QUALITY DATA

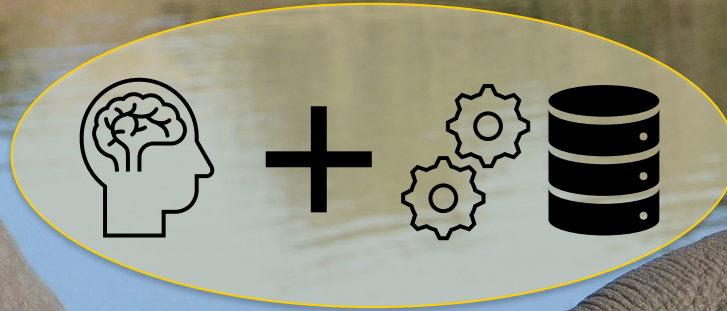
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CHRISTIN SEIFERT, SHREYASI PATHAK



AI in Health dream

Quality



Costs



Clinical decision making
Diagnosis & Treatment



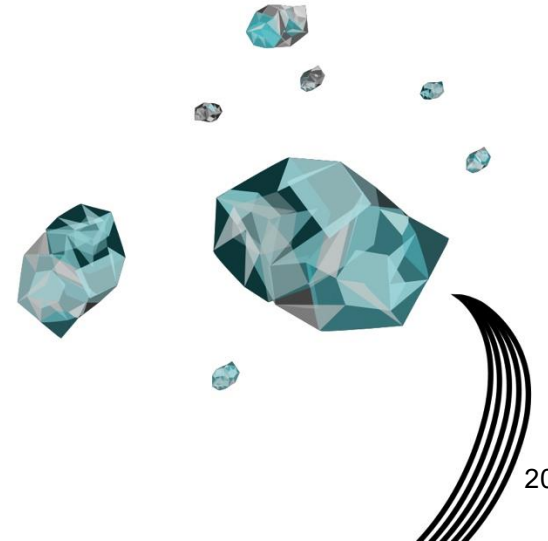
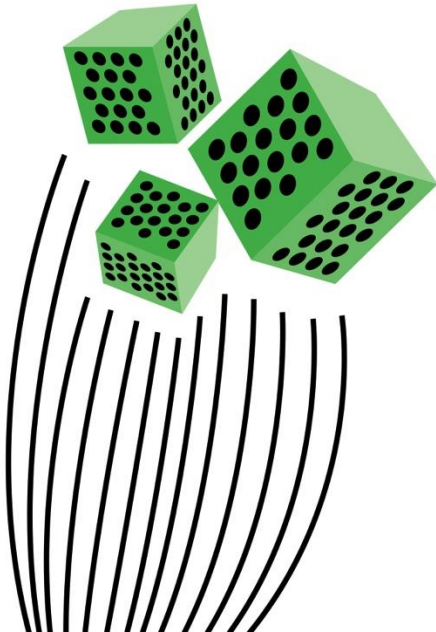
“Van een koude kermis
thuiskomen”

The struggle
for high quality data

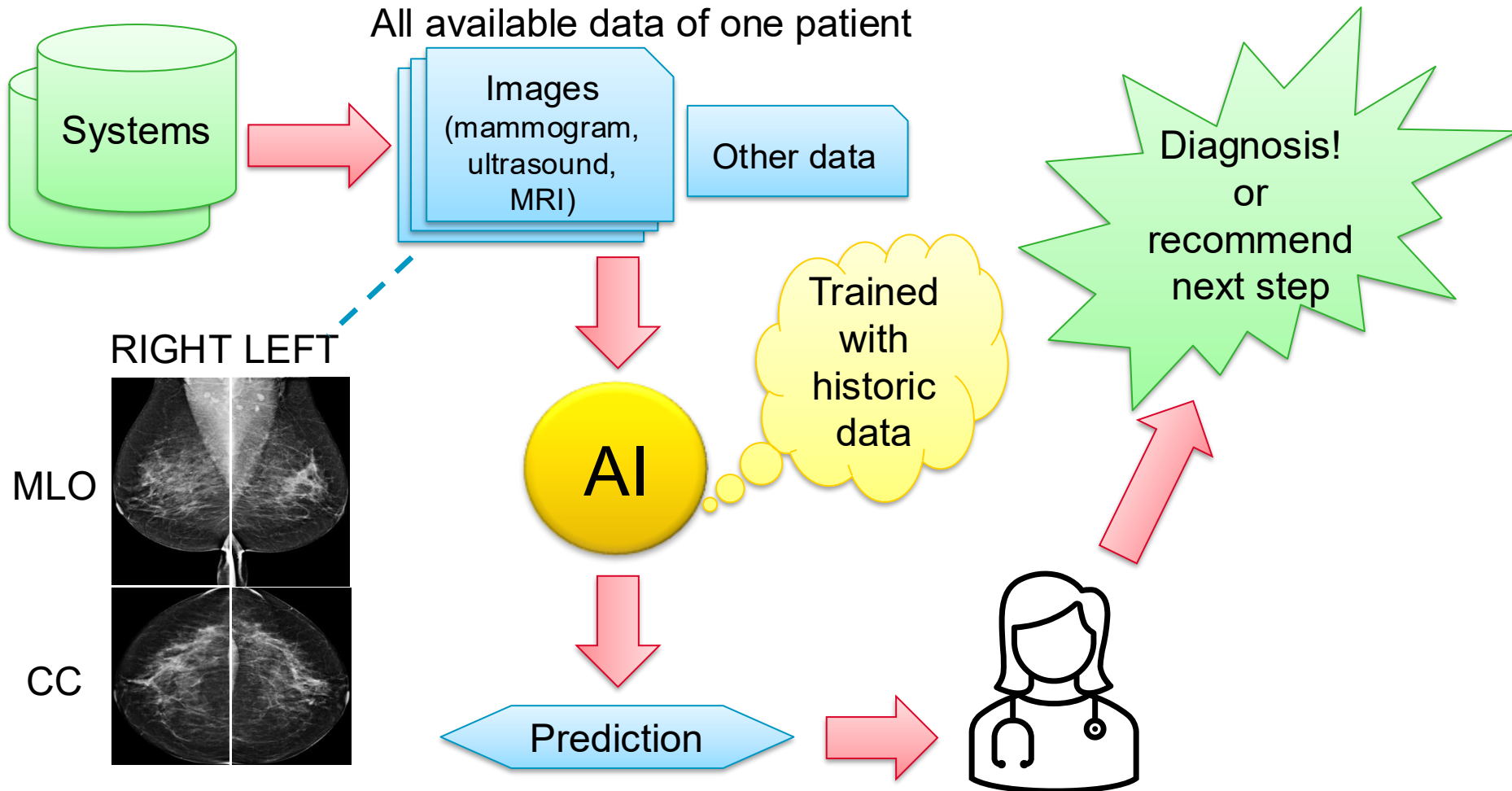
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A REPRESENTATIVE STORY ... OF A STRUGGLE

PHD PROJECT "SHREYASI PATHAK"



AIM: AI FOR BREAST CANCER DIAGNOSTICS



TRAINING DATA NEEDED

Labels

- Malignant vs. benign
- Finer: BIRADS 0..6

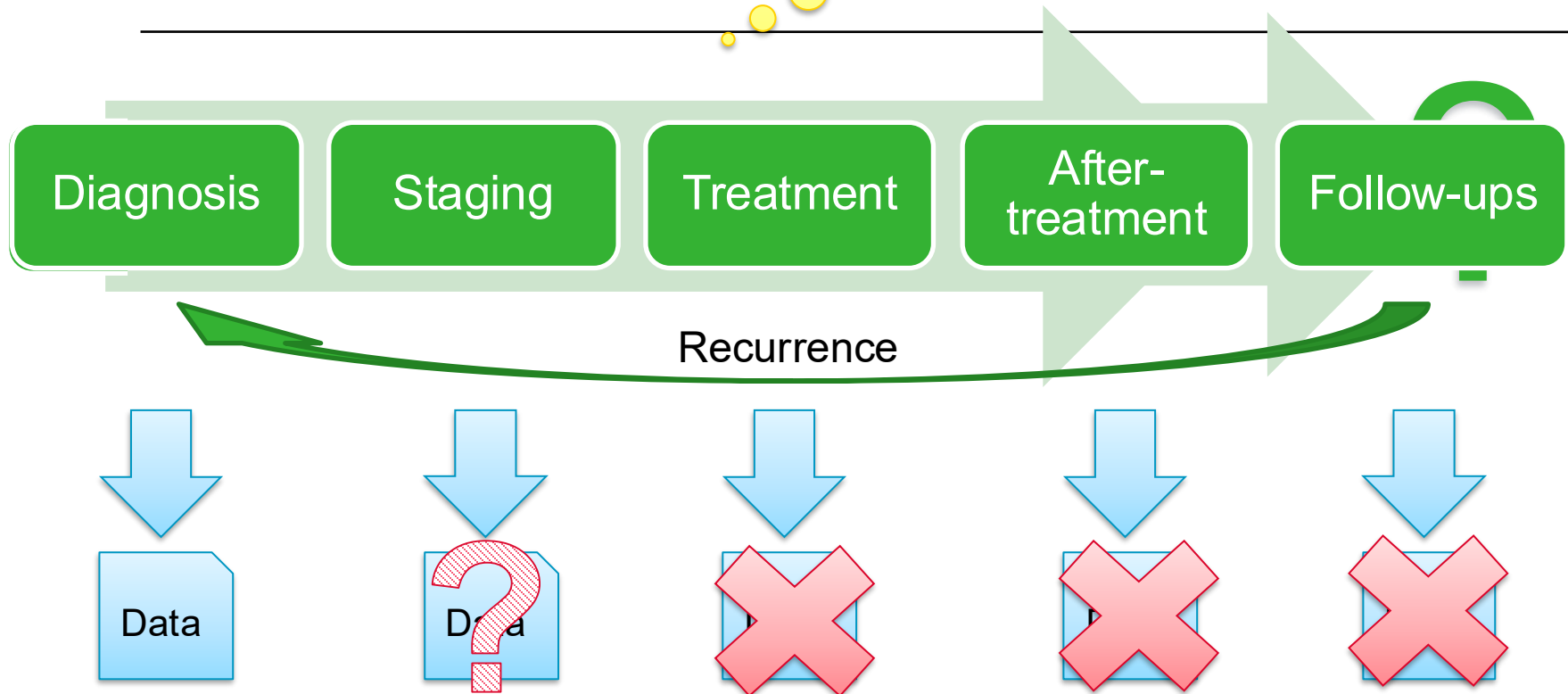
What data to select

- Focus on mammograms -> collect all of them
- Only diagnosis (no data after treatment)
- Only patients that have a final diagnosis

Fairly straightforward, right?
WRONG!

PATHWAYS

Even after removing follow-up pathways,
multiple pathways per patient remained.
What were those? Keep them?



No life cycle field

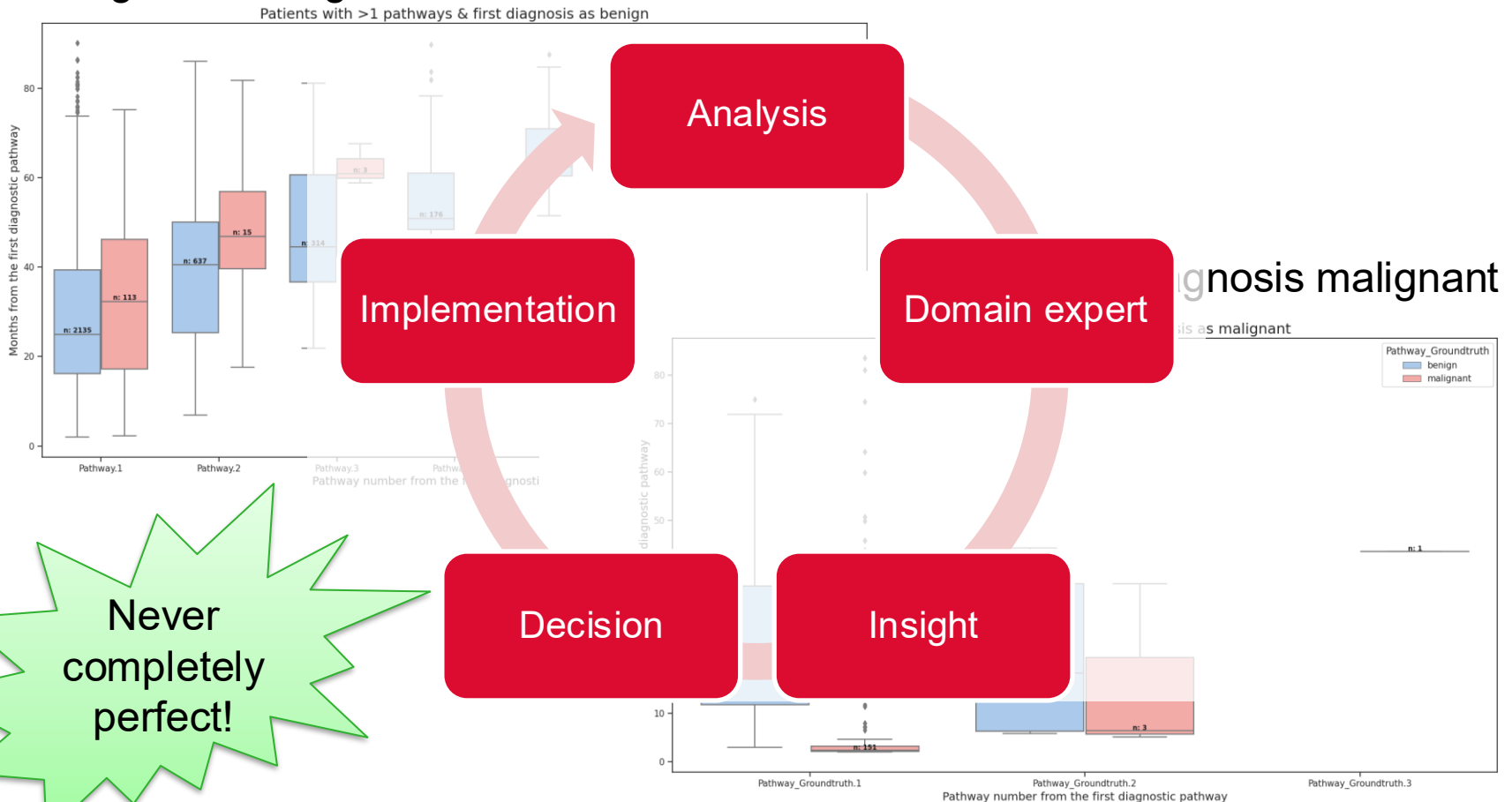
Vital facts buried in texts

How to recognize recurrence? Contradictory information

MULTIPLE PATHWAYS PER PATIENT

ADDITIONAL ANALYSIS TO MAKE INFORMED DECISION

First diagnosis benign



LABELS

OMG the problems we had
with establishing good
quality labels!

Problem 1: label per case, not per view

- If a case is malignant, the malignancy is not present in every view (e.g., only in left or right breast)

Problem 2: not always all views present

- We cannot just choose those cases with all views
 - Would render ML model useless in practice!
(in research people almost always do this 🤖 😬)

1/2a: no ML method for dealing with Problem 1 & 2

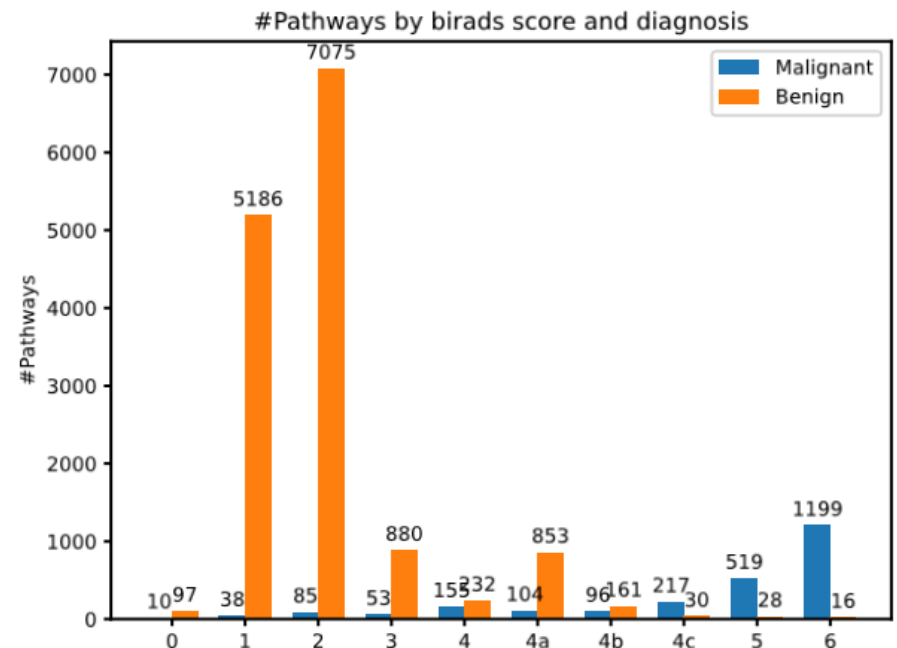
- We ended up researching ML method for classification of views with possibly views missing and case-based labels

LABELS

ONE WOULD THINK THAT BENIGN / MALIGNANT WOULD BE EASY

Problem 3: How to get reliable labels in the first place

- BIRAD scores and benign/malignant in conclusions of (pathology) reports => we need NLP to extract it
 - Multiple sources
=> inconsistency
 - More discussion with domain experts, re-review of individual cases and more analytics
- We ended up using (financial) DBC codes



MUCH MUCH MORE

Images

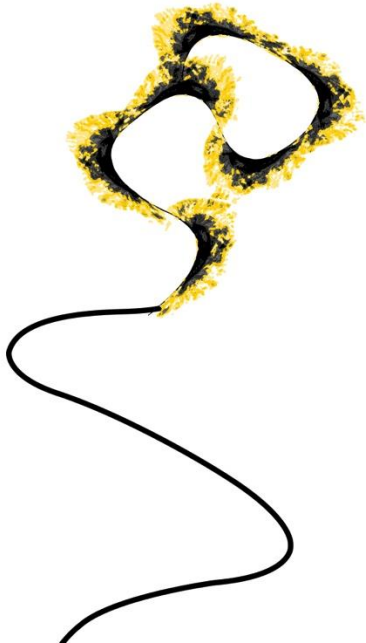
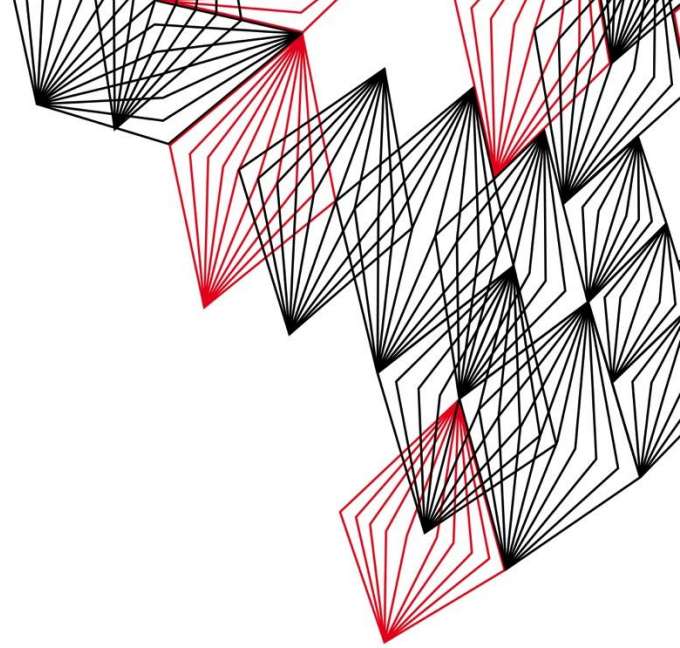
- Removing labels in the images
- Largest view of breast with highest resolution
- Increasing contrast

Anonimization + secure transfer from ZGT hospital (NL) to Essen Medical Computing Platform (EMCP; GER)

All this data collection, structuring,
and data quality improvement
took more than a year ...

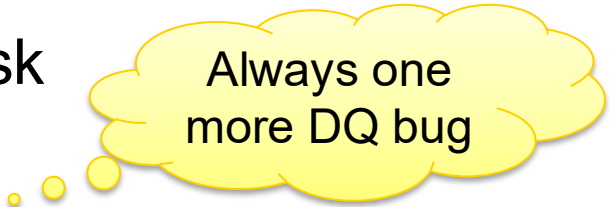
REFLECTION

WHY IS THIS SUCH A STRUGGLE?
WHAT CAN WE LEARN FROM THIS?



RECOMMENDATIONS FOR DATA SCIENTISTS

- Work with real-world data
- Do not oversimplify the ML task
- Do not exclude patients
- Check, Analyze, Check again
- Involve domain experts
- Expect to work 80% of your time on data preparation
- Document your data prep actions
- Be transparent and formulate conclusions carefully



Always one more DQ bug

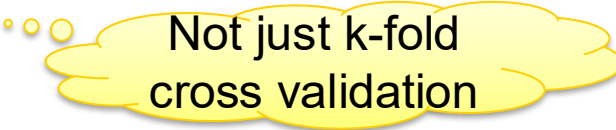
OPEN PROBLEMS THAT NEED RESEARCH



My main
research topic

AI-centric

- How to meaningfully use imperfect data
- Explainable AI
- How to properly validate ML models and AI systems



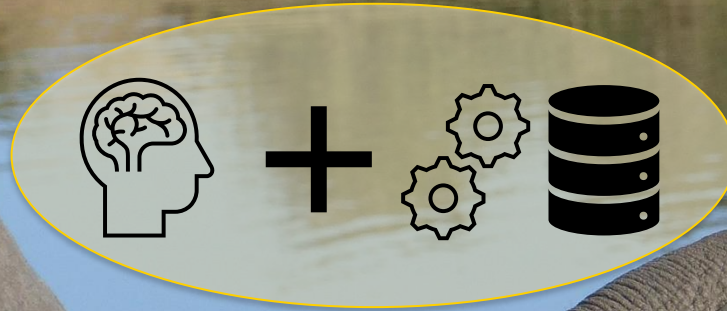
Not just k-fold
cross validation

Around the AI

- Redesign Data / IT software & enterprise architecture
 - AI enters primary processes of diagnosis & treatment decision making with new requirements
- Design AI as a socio-technical system
- Design of AI & Clinician interaction & workflow
- AI & Clinician collaboration (social sciences)

AI in Health dream

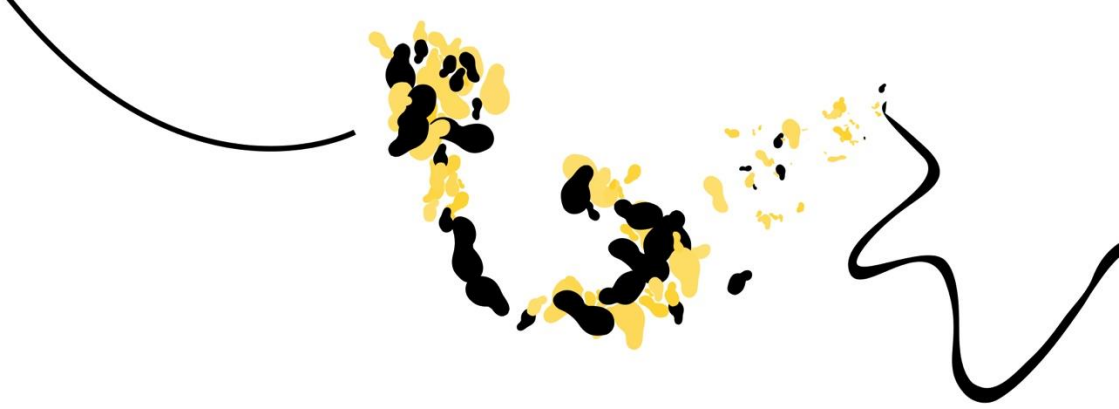
Quality



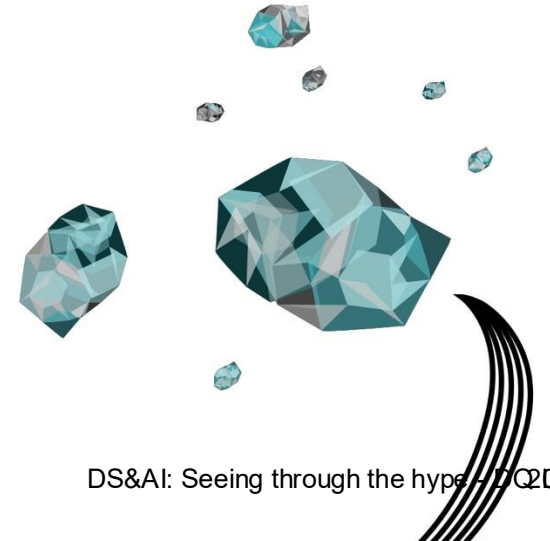
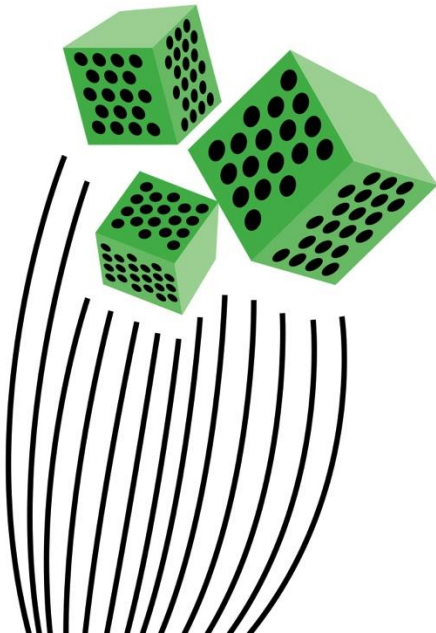
Costs



Clinical decision making
Diagnosis & Treatment



WHAT IS DATA QUALITY?



DEFINITION

“Data quality = Fit for use”

Use

Use = Purpose

- By customers
- In operations, decision-making, planning, etc.

Different purpose ->
different requirements

Fit for use

- Free of defects
- Possesses the features needed to complete the operation, decision, plan, etc.
- Exactly the right info, in the right place at the right time

Data science & AI
is a different purpose!

DATA USE NEEDS DOMAIN AND DATA UNDERSTANDING

A story about a mishap by a water authority ...

- Project would impact residents in the area
- They wanted to send them a letter announcing it
- How to get the addresses of these people?
 - Idea: all parcel owners withing a certain radius (we have that data anyway)
- Thus they sent many letters to dead people sending families into extra distress
 - ❖ Data on parcel ownership is updated only after inheritance procedures have been completed, which may sometimes take years ...

WHAT IS DATA?

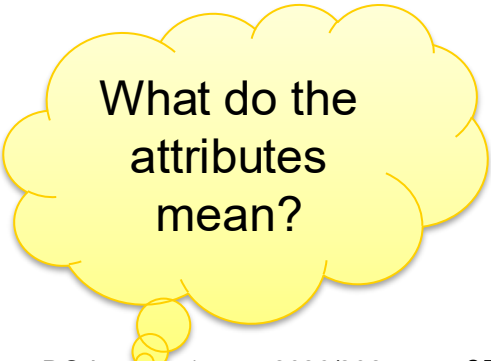
Data = Data model + Data values

Data model = database schema

- But there is so much data not in databases
- For example, spreadsheets

When talking about data quality, not only think about the quality of the data values, but also about

Documentation



What do the
attributes
mean?

DATA QUALITY DIMENSIONS

SOURCE: TABLE 2 ON PAGE 23 OF THE BOOK

None of these are measurable. For each there are one or more metrics

Conceptual view / Data model	Data values
Appropriate use Areas covered Attribute granularity Clear definition Comprehensiveness Essentialness Flexibility Homogeneity Identifiability Naturalness Obtainability Precision of domains Relevancy Robustness Semantic consistency Sources Structural consistency	Accuracy Completeness Consistency Timeliness Appropriateness
	Presentation quality
	Ease of interpretation Formats Format precision Flexibility Handling of null values Language Portability Representation Consistency Use of storage

PROPERTIES OF DATA THAT COMPLICATE ITS QUALITY

SOURCE: TABLE 5 ON PAGE 29

- Data multiplies
- Data are more complex than they appear
- Data are subtle and nuanced. They have become the organization's lingua franca
- Data create more value when they are “on the move”
- Data are organic
- Data can be digitized
- Data are the means by which organizations encode knowledge. They are meta-assets
- Data are intangible
- Each organization's data are uniquely its own

CLASSIFICATION OF DQ PROBLEMS

SOURCE: TABLE 1 ON PAGE 79

	Information perspective	User perspective
Context-independent	Spelling error Missing data Duplicate data Incorrect value Inconsistent data format Outdated data Incomplete data format Syntax violation Unique value violation Violation of integrity constraints Text formatting	The information is inaccessible The information is insecure The information is hardly retrievable The information is difficult to aggregate Errors in the information transformation
Context dependent	See next slide	

CLASSIFICATION OF DQ PROBLEMS

SOURCE: TABLE 1 ON PAGE 79

	Information perspective	User perspective
Context-independent	See previous slide	
Context dependent	- Violation of domain constraint - Violation of organization's business rules - Violation of company and government regulations - Violation of constraints provided by the database administrator	- The information is not based on fact - The information is of doubtful credibility - The information is irrelevant to the work - The information consists of inconsistent meanings - The information is hard to manipulate - The information is hard to understand

MINUTE DQ PROBLEMS CAN HAVE SEVERE IMPACTS

[HTTPS://TWITTER.COM/LIAMOSAUR/STATUS/1296305262144364544](https://twitter.com/liamosaur/status/1296305262144364544)

In 2019, "nathanwright120" edited Open Street Map

- Erroneous value 212 instead of 2
for one attribute of an object (#floors of a building)
of one of the billions of objects (building in Melbourne)
 - All other edits of user seemed fine, so probably typo
 - Error corrected by another user
-

But in the mean-time, Microsoft
used the Open Street Map data
for their Flight Simulator 2020

residential fawkner detail

Edited about 1 year ago by [nathanwright120](#)

Version #1 · Changeset [#73218100](#)

Tags

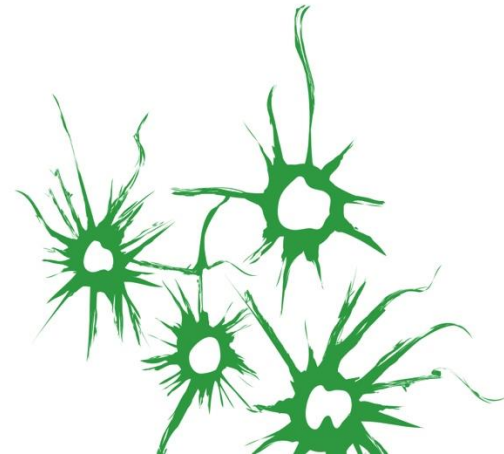
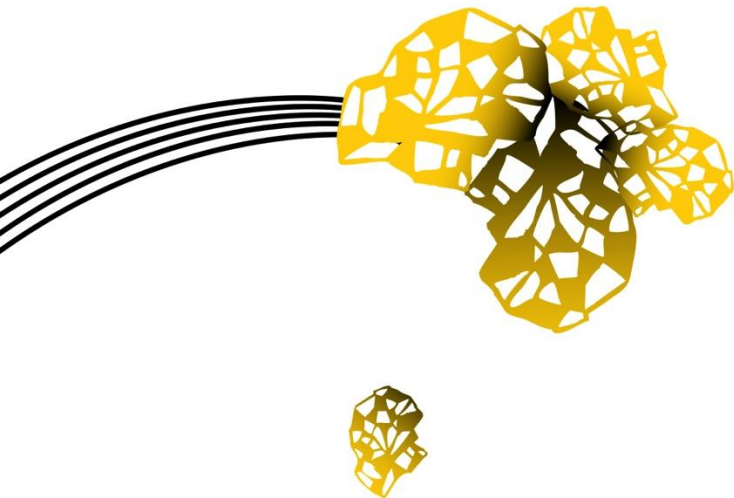
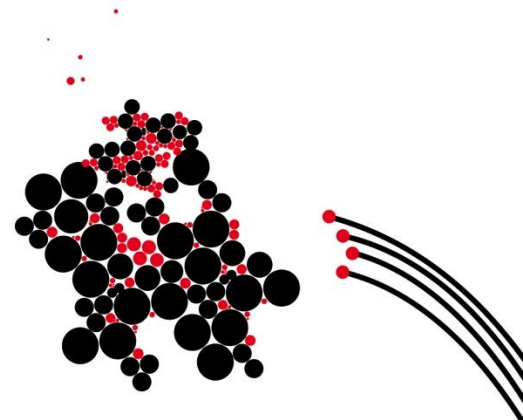
building	house
building:levels	212

MINUTE DQ PROBLEMS CAN HAVE SEVERE IMPACTS

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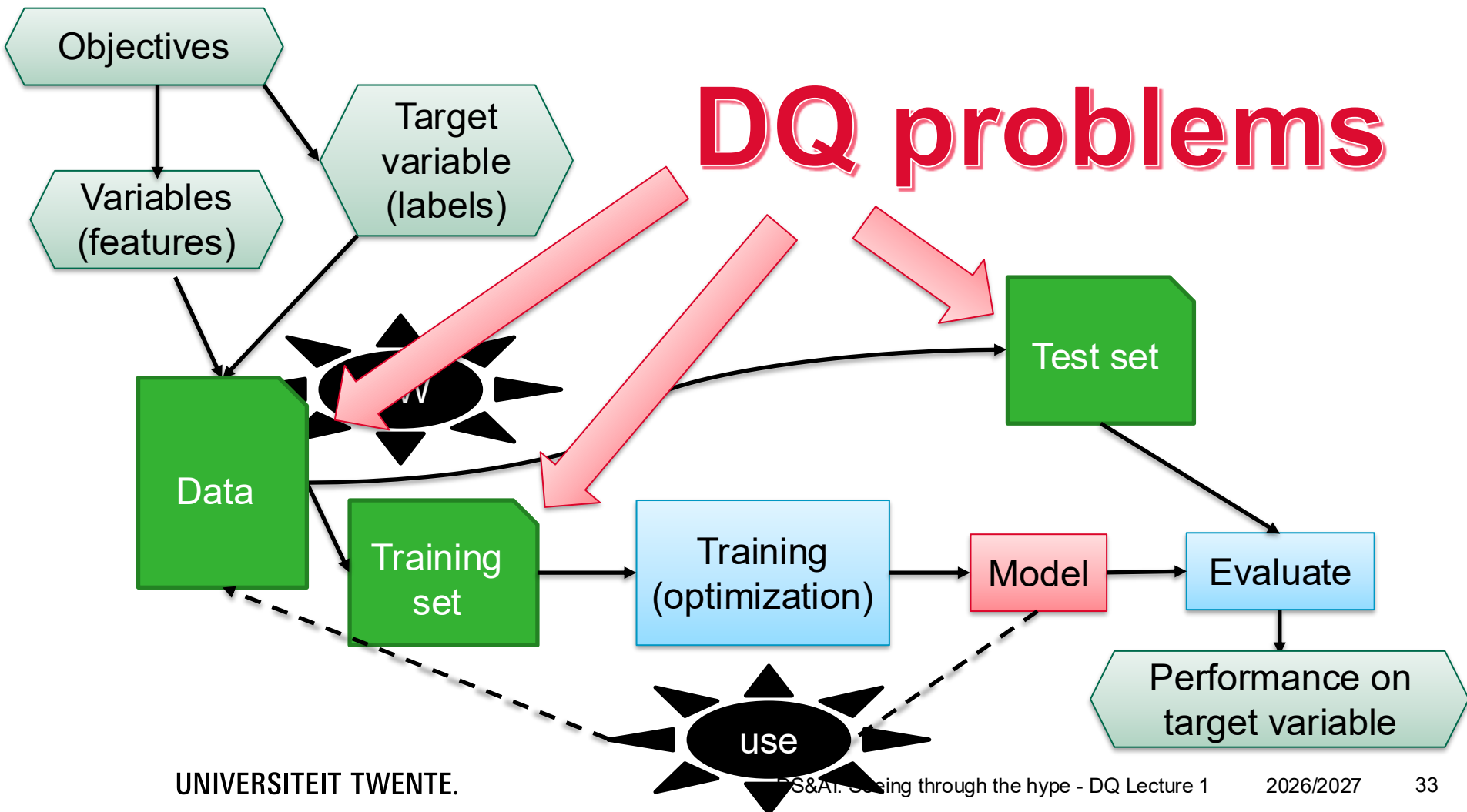


EFFECT OF DATA QUALITY ON MACHINE LEARNING PERFORMANCE



WHAT IS ML REALLY?

WHAT DOES IT REALLY DO?



HOW MUCH DATA CLEANING IS NEEDED?

Quick and
dirty or
thorough?

How does it
affect results?

What is good
enough?

Never
perfect!

- A data scientist should know ***and tell you*** about the deficiencies in the data and the results

LAB 1

Goal

- Experiment with data quality problems
 - of various kinds
 - using various machine learning methods
 - under various conditions
- To see what the effect is of these DQ problems on the performance of the resulting ML model

AI allowed for Lab 1

LAB 1

Read the instructions
in the introduction
carefully

You need **not do all** exercises!

- Experiments with classification and/or regression tasks
- Each exercise is either an *experiment* or an *expansion*
- You can choose a combination of one or more experiments with zero or more expansions yourself
- They are marked Easy, Regular, and Advanced
- Points and max points are indicated

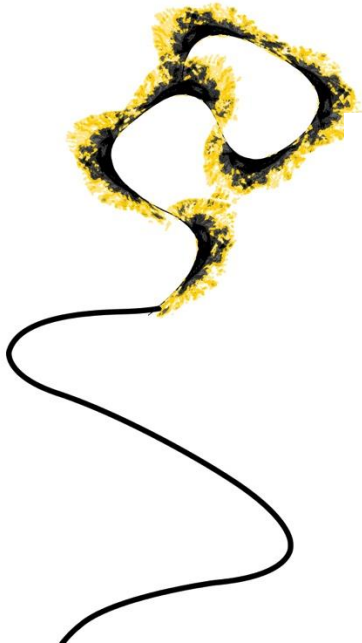
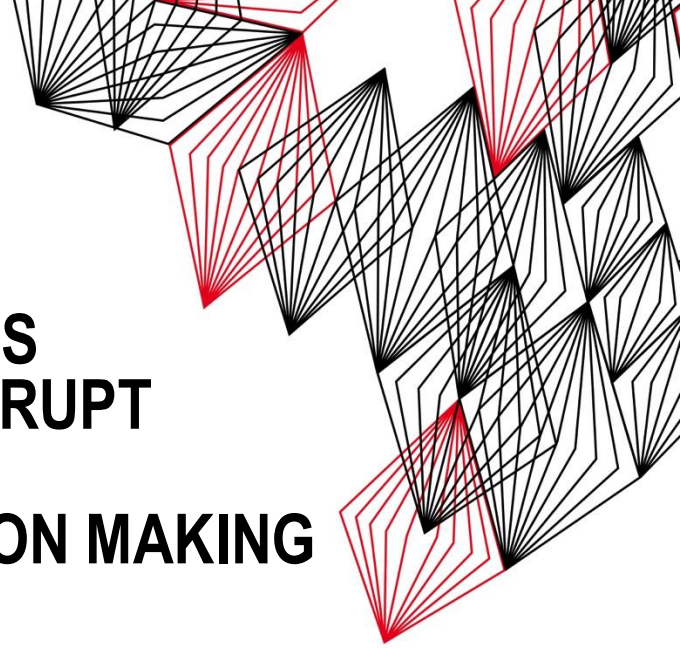
➤ And you need to write a reflection report (± 1 A4)

**DATA-DRIVEN DECISION MAKING IS
MORE OBJECTIVE AND LESS CORRUPT**

**BUT CAN ONE INFLUENCE DECISION MAKING
BY MANIPULATING DATA?**

Of course!

**Yet another reason
for a critical attitude**



DATA CAN BE MANIPULATED

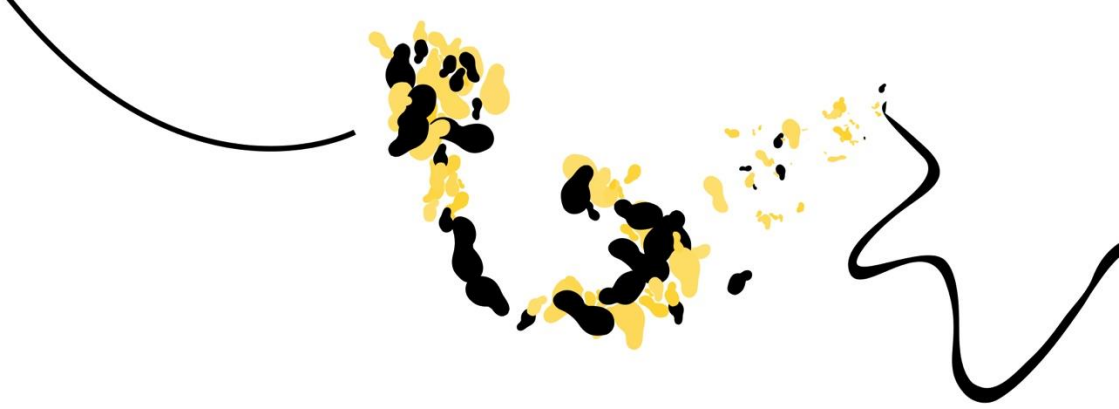
VALUE-TARGETED MANIPULATION

How to get parliament to approve of such a raise?

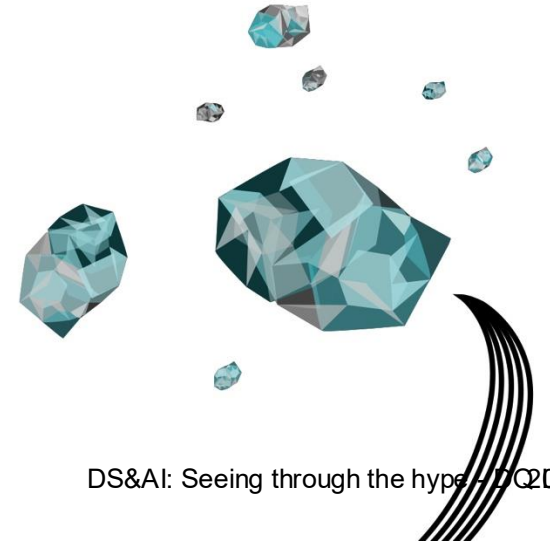
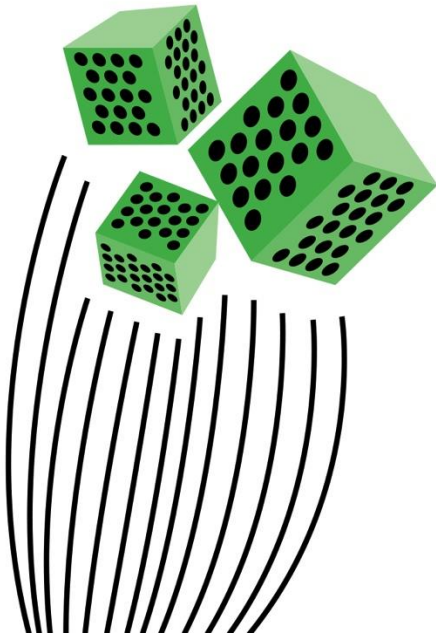
Head of civil service (CS) promised large pay raise to civil servants, but parliament voted themselves small one

1. 1000-page report; 1-page summary with main metrics
 - Avg salary increase and Total cost of CS pay
2. Increase bonuses everyone gets such as “London allowance”; these don’t count as salary
3. Reduce the CS to keep total cost down
 - Don’t call museum care takers and such “civil servant” anymore

See how we play with semantics to influence metrics?



WHAT IS SYMBOLIC REASONING?



KNOWLEDGE REPRESENTATION

Explicit representation of information about the world using symbols and composite expressions, in a form that can be used by a computer to reason and solve problems.

- A relational database is a knowledge representation
- But also
 - On(Bread, Table).
Holding(Woman, Milkjug)
- “Symbols” carry “Meaning”
- “Representation” refers to something in the real world



WHAT IS A BIRD?

- **How** do you know if something is a bird or not?
- Description logic representation of the concept “Bird”
 - $\text{Bird} \equiv \text{Animal} \sqcap \exists \text{hasCovering.Feather} \sqcap \exists \text{hasPart.Beak} \sqcap \exists \text{reproducesVia.Egg}$



WHAT IS A BIRD?

- $\text{Bird} \equiv \text{Animal} \sqcap \exists \text{hasCovering.Feather} \sqcap \exists \text{hasPart.Beak} \sqcap \exists \text{reproducesVia.Egg}$
- Counter examples?
 - Complies but is not a bird
 - Is a bird but doesn't comply



WHAT IS A BIRD?

- **When** does something stop being a bird?
- Why do we approach bird detection with ML nowadays? What do we loose?



REASONING

Reasoning: process of deriving **new information** and solving problems in a **logically sound way** on the basis of explicitly represented knowledge

- **Symbolic reasoning:** algorithms that do inference by manipulating symbols and expressions according to formal, well-defined rules
 - $\text{On}(\text{Bread}, \text{Table}). \text{In}(\text{Table}, \text{Room}).$
 $\text{On}(X, Y), \text{In}(Y, R) \Rightarrow \text{In}(X, R).$
We infer: $\text{In}(\text{Bread}, \text{Room})$
- **Neural reasoning:** inference by neural networks that represent knowledge as distributed patterns of weighted connections learned from data
 - $\text{Picture} \Rightarrow \text{Bread}:80\%; \text{Table}:91\%; \text{Room}:68\%$

SYMBOLIC VS NEURAL REASONING

Dimension	Symbolic	Neural (ML models)
Knowledge repres.	Explicit symbols/rules	Implicit, distributed weights
Inference	Formal logical derivation	Statistical pattern matching
Interpretability	High (traceable)	Low (requires XAI methods)
Learning	Typically hand-engineered	Learned from data
Handling of uncertainty/noise	Weak (needs extensions, e.g., probabilistic logic)	Strong (natively handles noisy input)
Compositional/systematic generalization	Strong	Weak
Scalability to unstructured data	Weak	Strong

NEURO + SYMBOLIC REASONING

- Neuro (ML): fallable and unpredictable
- Symbolic: deterministic, interpretable, efficient, auditable

ML → Symbolic: Symbolic inference on neural output

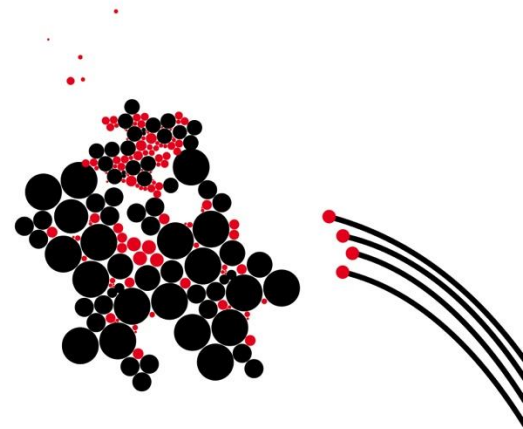
- Classification: Neuro inference → class (= symbol)
- GenAI evaluates per rubric → grade (= symbol)
Final grade: Weighted sum (symbolic reasoning)

Symbolic → ML: Neural approximation of symbolic inference

- Drug-target interaction prediction
Symbolic: 3D simulation of molecular docking
Neural: Predict interaction (classification)
Symbolic algorithm used to general training data

Hybrid both ways

- Chess engine: move generation, endgame (symbolic), position evaluation, search guidance (neural)



CONCLUSION

MOTIVATION

WHAT IS DATA QUALITY

WHAT IS NEURO-SYMBOLIC AI

EFFECT OF DQ ON ML

PROPERTIES OF NEURO & SYMBOLIC

COMBINE ML WITH SYM. REASONING

